

Flare Activity and Rotation Rates Among TESS Main Sequence Wide Binaries

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Flare Detection with AltaiPony Python Algorithm

- The AltaiPony (AP) Python-based algorithm detrends, stores, and analyzes the long and short cadence TESS light curves to search for flares [1] [2]
- AP has a detrending algorithm that removes periodic and noise signals from light curves (Figure 1)
- The AP flare program has several limitations:
 - Blends – user must implement a separate algorithm to work with these
 - Pulsating stars – currently cannot distinguish pulsations from flares
 - Noise and incomplete detrending may result in false flare detections
- Over 1956 pairs of wide, noninteracting, TESS binaries were analyzed for flaring tendencies, with over 10% yielding some sort of flare signal
- ~200 flaring candidates were visibly non-contaminated

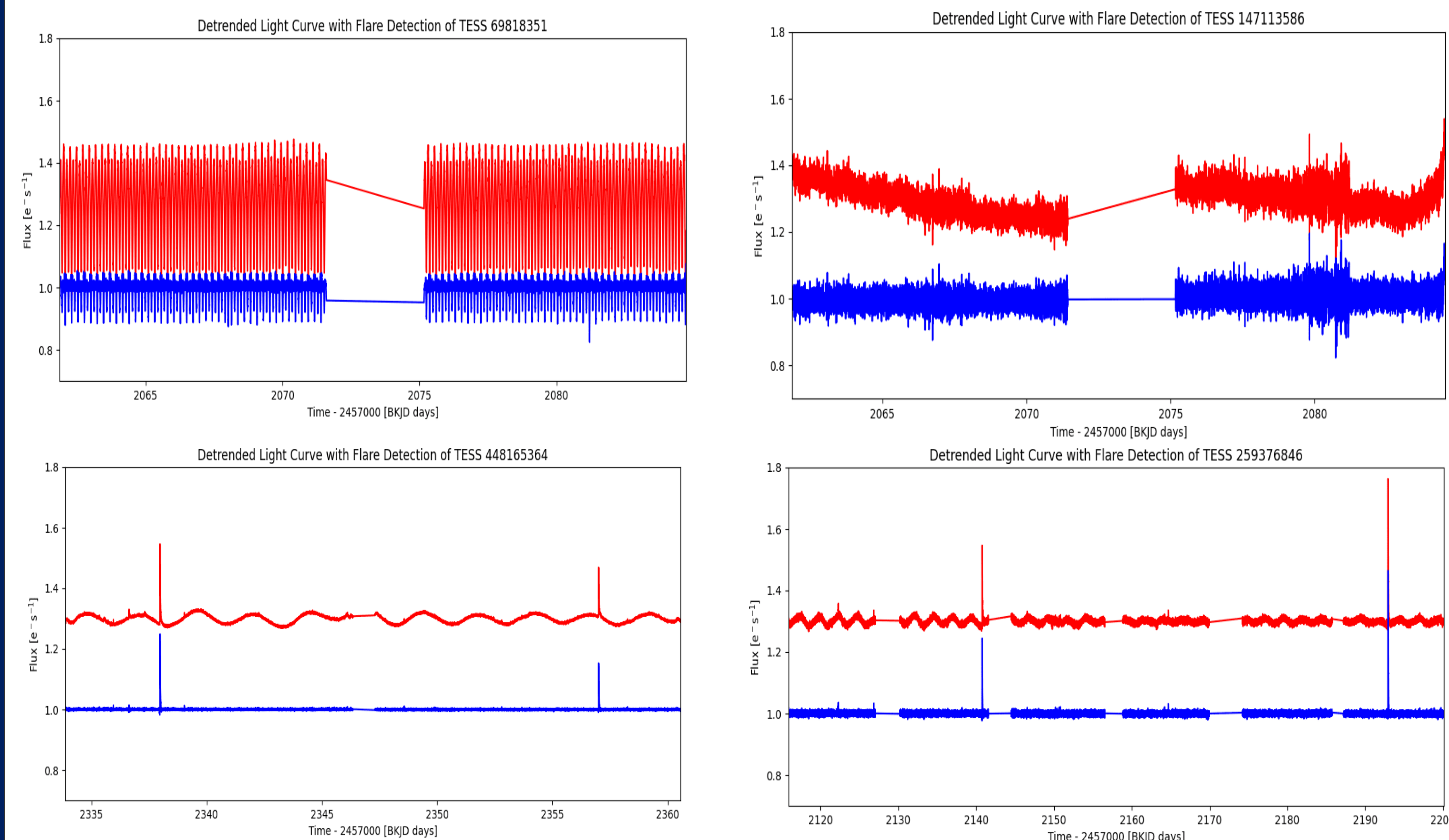


Figure 1 – Four plots representing some of the limitations of AP and various probable flare candidate examples. All four plots have the normalized detrended flux centered at 1.0 and normalized flux centered at 1.3. The top left plot is a pulsating star with over 100 detected flares, the top right is an example of instrumental errors causing false flare detections, and the bottom two plots are potential flare candidates.

Next Steps

- Extend flare detection work to Kepler Mission data, which exceeds total time coverage by TESS by almost an order of magnitude
- Determine whether activity cycles can be detected

References

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- [6] Wright et al. (2011). THE STELLAR-ACTIVITY-ROTATION RELATIONSHIP AND THE EVOLUTION OF STELLAR DYNAMOS. *The Astrophysical Journal*, 743(48), 16. <https://doi.org/10.1088/0004-637x/743/1/48>

ABSTRACT

We present a study of activity among lower main sequence stars using Transiting Exoplanet Survey Satellite (TESS) observations from which the rotation periods have been determined. This study uses a Python-based program called AltaiPony to detect flaring activity among 3912 wide, non-interacting binaries. We examine the correlation between stellar activity and stellar parameters such as rotation rate, Rossby number, luminosity, and mass. The well-known relation between stellar rotation and age, known as Gyrochronology, predicts that as a star ages, its rotation period increases. We found that flare activity and intensity decrease as stellar rotation periods increase, supporting the notion that flare activity diminishes with age.

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Stellar Activity Relationships

- The Rossby number (R_o) is defined as the ratio of rotation period and the convective turnover time [6]; it is more closely related to a star's activity level than rotation period alone
- Figure 2 displays the correlations we found between stellar activity, mass, luminosity, and the rotation periods (determined by the Lomb-Scargle periodogram analysis)
- Flare frequencies in these images were underestimated in value, but the trends and patterns are not affected
- Each plot below shows a decrease in flare frequency and flare intensity as rotation periods increase (aging stars)

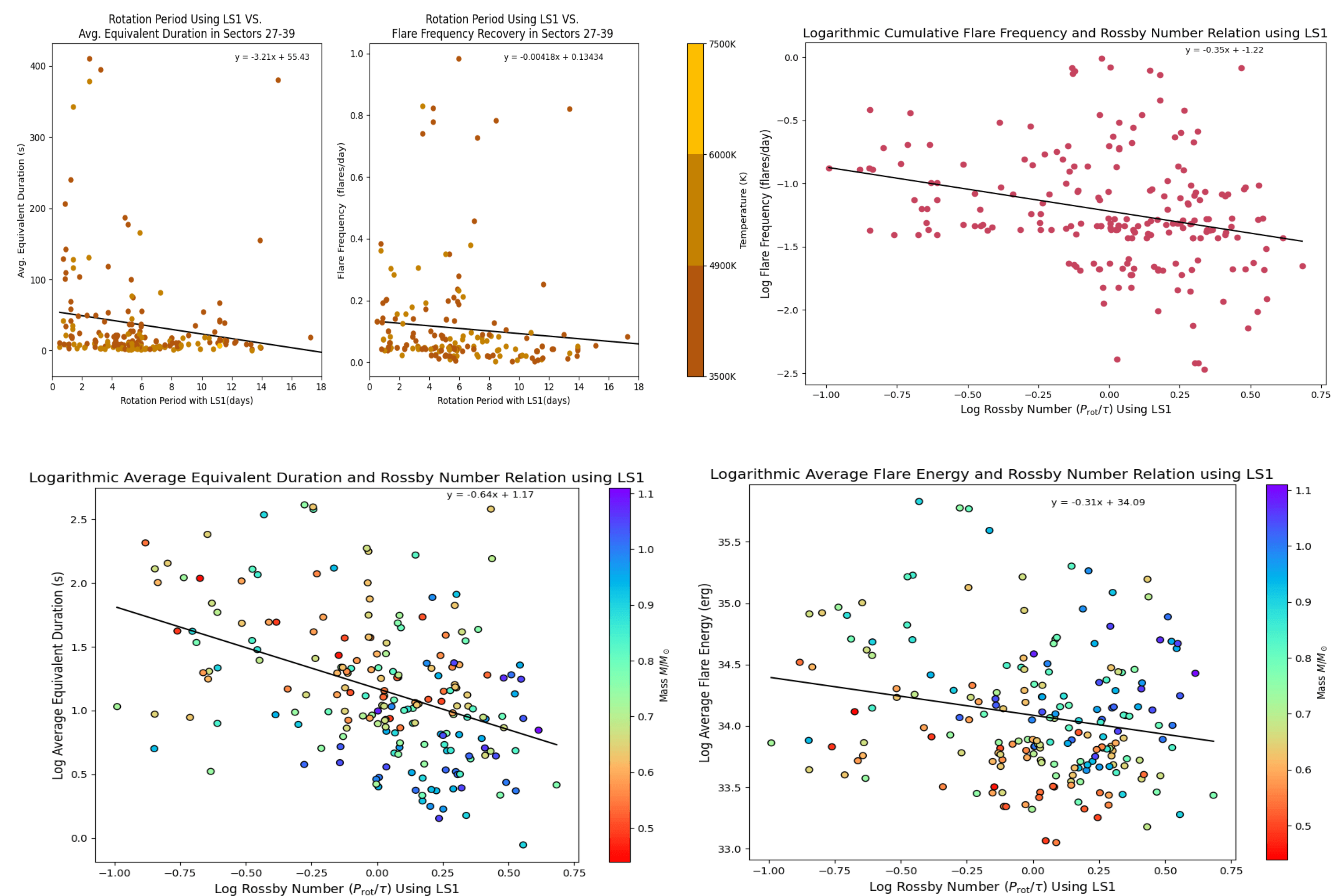


Figure 2 - Four plots showing different relationships between various parameters (such as mass, temperature, Rossby number and rotation periods) and stellar flare activity. The top left plot contains a temperature color bar, and the bottom two plots contain mass color bars.